SIMATS ENGINEERING
ASSESSMENT TOOL 1

COURSE NAME: Artificial Intelligence   COURSE CODE: CSA17

COURSE OUTCOME COVERED

CO3: *Analyze knowledge representation and reasoning using propositional logic, FOL, inference, resolution, chaining methods, knowledge representation to perform logical reasoning for drawing valid conclusions and decision making. (BL4)*

Assessment Tool Weightage: 50%

Time Duration: 1 Hour

QUESTIONS: 4   **Total Marks:40**

Annexure A

CASE STUDY

Code of Conduct:

*I **MUKKARAM** (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:
MUKKARAM

Annexure A

CASE STUDY

Case Study 1 — Smart Medical Diagnosis System

A hospital is designing a rule-based Logical Agent for preliminary patient diagnosis. The system's knowledge base contains:

I. If a patient has Fever AND Cough → Possible Flu

II. If a patient has Fever AND Rash → Possible Measles

III. If a patient has Cough AND Breathlessness → Possible Pneumonia

IV. Facts about Patient A: Fever = True, Cough = True, Breathlessness = True

(a) Represent the above rules and facts precisely using Propositional Logic symbols. Define each symbol used.

(b) Apply Forward Chaining on the knowledge base to derive all possible diagnoses for Patient A. Show each inference step with the rule fired and fact derived.

(c) The goal is to confirm 'Patient A has Pneumonia.' Apply Backward Chaining and draw the goal reduction tree to verify this goal.

Case Study 2 — Autonomous Traffic Management Agent

A smart city traffic controller uses a FOL-based knowledge base to manage emergency vehicle routing:

I. ∀x: Vehicle(x) ∧ Emergency Type(x) → ClearPath(x) [Rule 1]
II. ∀x: ClearPath(x) → Green Signal(x) ∧ ¬Red Signal(x) [Rule 2]
III. ∀x ∀y: Vehicle(x) ∧ Behind (x, y) ∧ Green Signal(y) → Proceed(x)[Rule 3]
IV. Facts: Vehicle (Ambulance), Emergency Type (Ambulance), Vehicle (Car A), Behind (Car A, Ambulance)

(a) Apply Unification to match Rule 1 with the given facts. Show the complete substitution set θ (MGU) at each step.

(b) Using the Resolution method, prove that 'Proceed (Car A)' is true. Convert relevant clauses to CNF and show each resolution step.

(c) Analyze whether the current FOL knowledge base is sufficient to handle two simultaneous emergency vehicles. Identify what additional rules or facts are needed and justify logically.

Case Study 3 — Agricultural Expert Reasoning System

An agricultural advisory logical agent operates with the following propositional knowledge base (KB):

1) P1: Soil Dry → Irrigation Needed
2) P2: Irrigation Needed ∧ Crop Wheat → Apply Drip Method
3) P3: ¬Apply Drip Method ∧ Crop Wheat → Crop at Risk
4) Facts: Soil Dry = True, Crop Wheat = True

(a) Convert all sentences P1, P2, P3 and the facts into Conjunctive Normal Form (CNF). Show each conversion step (eliminate implication → move ¬ inward → distribute).

(b) Using Resolution Refutation, prove the theorem 'Apply Drip Method'. Negate the goal, combine with the KB in CNF, and derive the empty clause □. Draw the resolution proof tree.

(c) If the fact 'Soil Dry' is removed from the KB, analyze step-by-step how logical conclusions change. Does 'Crop at Risk' still follow? Apply appropriate reasoning patterns to justify.

Case Study 4 — Wumpus World Knowledge Agent

An AI agent navigating the Wumpus World perceives: Stench at [1,2], Breeze at [1,1], Glitter at [2,2]. The agent uses a propositional KB with these rules:

1) Stench [1,2] → Wumpus is adjacent to cell [1,2]
2) Breeze [1,1] → Pit is adjacent to cell [1,1]
3) Glitter [x, y] → Gold is at [x, y]
4) ¬Wumpus [x, y] ∧ ¬Pit [x, y] → Safe [x, y]

(a) Construct the agent's belief state from its percepts. Apply Modus Ponens systematically to derive new knowledge facts. List all derived conclusions.

(b) Using Forward Chaining from the agent's percepts, identify the possible locations of the Wumpus and Pit. Show each reasoning step precisely.

(c) Evaluate whether the path [1,1] → [1,2] → [2,2] is safe for the agent to collect the gold. Use the knowledge derived in (a) and (b) to justify your conclusion with logical inference.

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Case	25	Thorough understanding; clearly identifies all key issues	Good understanding with most issues identified	Basic understanding; some issues missed	Poor understanding; problem not identified correctly
Application of Concepts	25	Effectively applies relevant concepts to analyze the case deeply	Applies appropriate concepts with minor gaps	Limited application of concepts	Fails to apply relevant concepts
Analysis	20	In-depth analysis with strong reasoning and insights	Good analysis with logical reasoning	Basic analysis with limited depth	Weak or no analysis; lacks reasoning
Solution Design	20	Innovative, feasible solutions with strong justification	Logical solutions with adequate justification	Basic solutions with weak justification	Incorrect or no solution provided
Clarity	10	Clear, well-structured, and easy to understand	Organized and understandable presentation	Limited clarity and structure	Poor clarity; difficult to understand

Case Study 1 — Smart Medical Diagnosis System

(a) Representation using Propositional Logic

Define the symbols

Let:

- **F** = Patient has Fever
- **C** = Patient has Cough
- **B** = Patient has Breathlessness
- **Flu** = Possible Flu
- **M** = Possible Measles
- **P** = Possible Pneumonia

Rules

Rule I: Fever AND Cough → Possible Flu

Rule II: Fever AND Rash → Possible Measles

Let **R** = Patient has Rash.

Rule III: Cough AND Breathlessness → Possible Pneumonia

Facts about Patient A

Therefore, the initial knowledge base is:

(b) Forward Chaining

Forward chaining starts with the **known facts** and repeatedly applies rules whose conditions are satisfied.

Initial facts

Step 1 — Apply Rule I

Rule:

Since:

- Fever = True
- Cough = True

Both conditions are satisfied.

Therefore:

Derived fact: Patient A may have Flu.

Step 2 — Apply Rule II

Rule:

We know:

But:

is **not given**.

Therefore, Rule II cannot fire.

No Measles conclusion.

Step 3 — Apply Rule III

Rule:

We know:

and

Therefore:

Derived fact: Patient A may have Pneumonia.

Final Forward-Chaining Conclusions

Measles cannot be derived because Rash is not known.

(c) Backward Chaining for Pneumonia

Goal

Backward chaining starts from the goal and searches for rules that can produce it.

Goal Reduction Tree

```
          Goal
      Pneumonia(A)
           |
      Rule III
```

```
              |
        Cough(A) AND
      Breathlessness(A)
          /        \
         /          \
   Cough (A) Breathlessness(A)
       ✓              ✓
```

Explanation
To prove:
we need:
Both are given as facts.
Therefore:
Final conclusion
Patient A has possible Pneumonia according to the rule-based knowledge base.

Case Study 2 — Autonomous Traffic Management Agent

Given FOL Knowledge Base
Rule 1
Rule 2
Rule 3
Facts
(a) Unification
We want to apply Rule 1:
Match with first fact
Rule condition:
Fact:
Therefore:
Match with second fact
Rule condition becomes:
Fact:
This matches successfully.
Therefore, the **Most General Unifier (MGU)** is:
Applying the substitution
Original conclusion:
After applying:
we get:
(b) Resolution Proof of Proceed (Car A)
We want to prove:
Using resolution, negate the goal:
Step 1 — Convert Rule 1 to CNF
Original:
Remove implication:
Therefore:
Step 2 — Convert Rule 2
Rule 2:
This gives two clauses:
Only C2 is required to prove Proceed.
Step 3 — Convert Rule 3
CNF:
Facts
Negated goal:
Resolution Step 1
Resolve C1 with C5:

and
Using:
we obtain:
Resolution Step 2
Resolve C10 with C6:
and:
Therefore:
Resolution Step 3
Use Rule 2:
with:
Therefore:
Resolution Step 4
Use Rule 3:
Substitute:
We get:
Using the known facts:
we derive:
Step 5 — Resolution with Negated Goal
We have:
and:
Therefore:
The empty clause is obtained.
Conclusion
Thus, **Car A can proceed behind the ambulance** according to the given knowledge base.
(c) Handling Two Simultaneous Emergency Vehicles
The current knowledge base can represent **multiple vehicles**, because the rules use variables such as
and
For example, suppose we have:
Rule 1 can derive:
and:
Rule 2 can then derive:
and:
However, a problem exists
The knowledge base does **not specify how the traffic controller should handle conflicts between two emergency vehicles**, such as:

- Which vehicle gets priority?
- What happens if they approach the same intersection?
- How should conflicting green signals be handled?
- What happens if they travel in different directions?

Additional rules needed
For example:
And:
We may also need:
Conclusion
The KB is **sufficient to identify multiple emergency vehicles**, but it is **not sufficient for conflict resolution and priority management** between simultaneous emergency vehicles.

Case Study 3 — Agricultural Expert Reasoning System

Given Knowledge Base
P1
P2
P3
Facts
(a) Convert into CNF
P1
Original:
Eliminate implication
Using:
Therefore:
This is already CNF.
P2
Original:
Eliminate implication:
Using De Morgan's law:
Therefore:
P3
Original:
Eliminate implication:
Move negation inward:
Therefore:
Facts in CNF
Fact 1:
Fact 2:
Final CNF
(b) Resolution Refutation to Prove Apply Drip Method
Goal
Negate the goal:
Add this to the KB.
Resolution Step 1
Use:
and:
Resolve:
Resolution Step 2
Use:
and:
Resolve:
Resolution Step 3
Use:
and:
Therefore:
Resolution Step 4
We added the negated goal:
Resolve:
with:
Therefore:
Resolution Proof Tree

```
Soil Dry
  |
  | C1
```

↓
Irrigation Needed
|
| C2
↓
¬Crop Wheat ∨ Apply Drip Method
|
| Crop Wheat
↓
Apply Drip Method
|
| ¬ Goal
↓
¬Apply Drip Method
|
↓
□

Therefore:

(c) Remove Soil Dry from the KB

Suppose:

is removed.

Remaining fact:

Step 1

P1 is:

Since **Soil Dry is no longer known**, we cannot derive:

Step 2

P2 requires:

Although:

we do not know:

Therefore:

Step 3 — Crop At Risk

P3 requires:

We know:

But we **do not know**:

Therefore:

Important logical point

Not being able to prove Apply Drip Method does **not** mean that ¬Apply Drip Method is true.

This is the principle of **open-world reasoning**.

Final conclusion

After removing Soil Dry:

- Irrigation Needed → **Cannot be derived**
- Apply Drip Method → **Cannot be derived**
- Crop At Risk → **Cannot be derived**

Case Study 4 — Wumpus World Knowledge Agent Given Percept

The agent perceives:

1. Stench at
2. Breeze at
3. Glitter at

(a) Construct the Belief State

Let:

- = Stench at [1,2]
- = Breeze at [1,1]

- = Glitter at [2,2]
- = Wumpus
- = Pit
- = Gold
- = Safe

Initial percept
Step 1 — Apply Rule 1
Rule:
Since:
is true:
Step 2 — Apply Rule 2
Rule:
Since:
is true:
Step 3 — Apply Rule 3
Rule:
Given:
Therefore:
The agent knows that gold is at.
Step 4 — Apply Rule 4
Rule:
However, the agent does not have explicit facts such as:
or:
Therefore, **Safe cells cannot be conclusively derived from Rule 4**.
Derived belief state
No definite Safe cell is derived from the given rules.
(b) Forward Chaining — Possible Wumpus and Pit Locations
Assume the standard Wumpus World uses **four-directional adjacency**: up, down, left and right.
Wumpus
We have:
Therefore:
The adjacent cells are:
Therefore, possible Wumpus locations are:
assuming these cells are within the board.
Pit
We have:
Therefore:
Adjacent cells are:
Therefore, possible Pit locations are:
Gold
We have:
Therefore:
Summary

Object	Possible locations
Wumpus	[1,1], [1,3], [2,2]
Pit	[1,2], [2,1]
Gold	[2,2]

(c) Is the Path [1,1] → [1,2] → [2,2] Safe?
The proposed path is:
[1,1] → [1,2] → [2,2]
The agent wants to reach:
to collect the gold.

Check Cell [1,1]
There is a Breeze at:
This tells us that a pit may be adjacent to [1,1].
However, the rules do not prove that [1,1] itself contains a pit.
So:
But it is also **not proven safe** under Rule 4.
Check Cell [1,2]
This is more dangerous.
We know:
and therefore, a pit could be at:
Also, there is:
which tells us a Wumpus is adjacent to [1,2].
Therefore, [1,2] has unresolved hazards.
Check Cell [2,2]
We know:
However, from the stench at [1,2], [2,2] is one possible location of the Wumpus.
Therefore:
is **not guaranteed to be safe** either.
Final conclusion
The path:
cannot be proven safe using the given knowledge base.
The main reason is that the KB only provides **possible hazard locations**, not enough negative facts to establish safety.
In particular:
could contain a pit, and:
could potentially contain the Wumpus.
Therefore, the agent should **not classify the complete path as safe** based on the current knowledge.